

Transformations of Quadratic Functions - Worksheet

MCR3U

SOLUTIONS

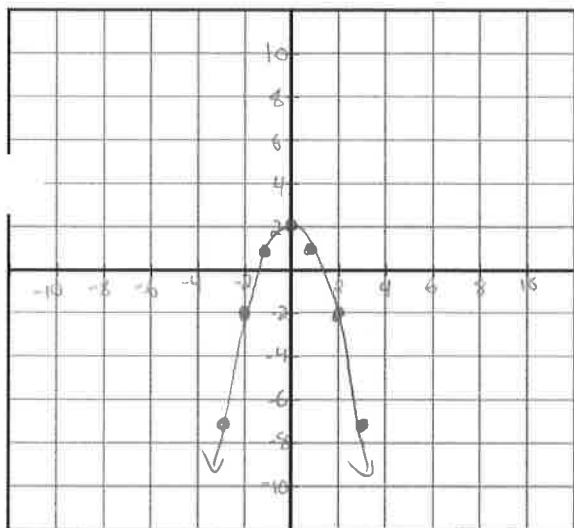
Key points for
 $y = x^2$

x	y
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9

1) For each of the following graphs:

- describe the transformations in order ($a \rightarrow k \rightarrow d \rightarrow c$)
- create a table of values for the transformed function
- graph the transformed function

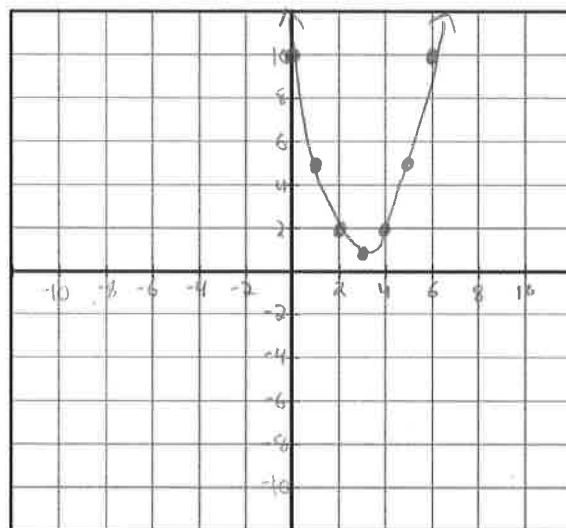
a) $y = -x^2 + 2$



- vertical reflection ($-y$)
- shift up 2 ($y+2$)

x	-y+2
-3	-7
-2	-2
-1	1
0	2
1	1
2	-2
3	-7

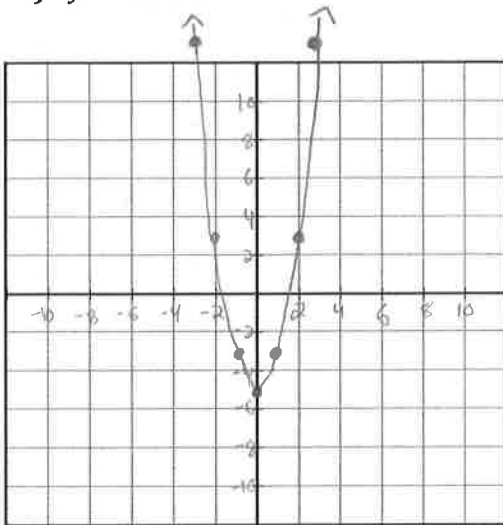
b) $y = (x - 3)^2 + 1$



- shift right 3 units ($x+3$)
- shift up 1 unit ($y+1$)

x+3	y+1
0	10
1	5
2	2
3	1
4	2
5	5
6	10

c) $y = 2x^2 - 5$

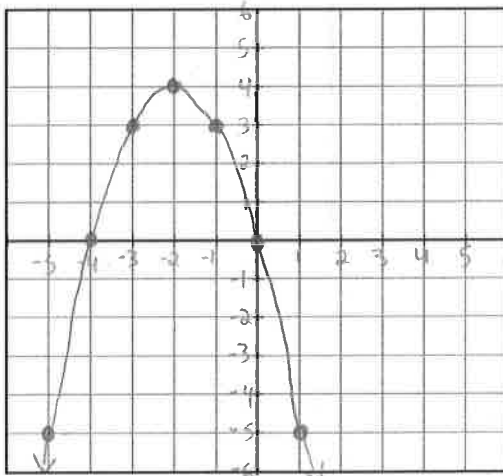


(-3, 13)
(-2, 3)
(-1, -3)
(0, -5)
(1, -3)
(2, 3)
(3, 13)

- 1) vertical stretch bafco 2 ($2y$)
- 2) shift down 5 units ($y-5$)

x	$2y-5$
-3	13
-2	3
-1	-3
0	-5
1	-3
2	3
3	13

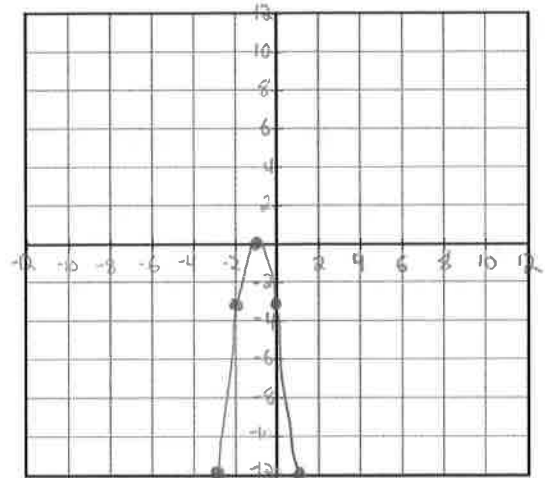
e) $y = -(x+2)^2 + 4$



- 1) vertical reflection ($-y$)
- 2) shift left 2 units ($x-2$)
- 3) shift up 4 units ($y+4$)

$x-2$	$-y+4$
-5	-5
-4	0
-3	3
-2	4
-1	3
0	0
1	-5

d) $y = -3(x+1)^2$

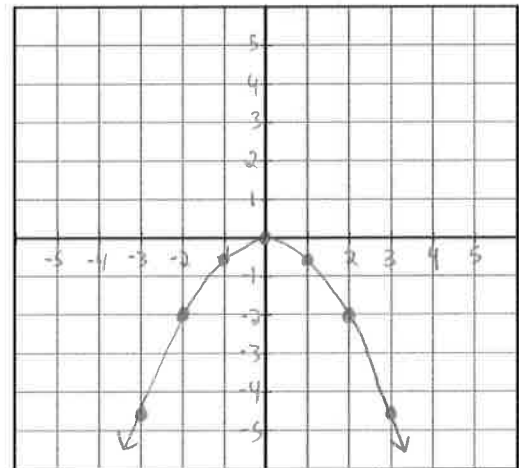


- 1) vertical stretch by 3 and vertical reflection ($-3y$)
- 2) phase shift left 1 unit ($x-1$)

$x-1$	$-3y$
-4	-12
-3	-3
-2	0
-1	0
0	-3
1	-12
2	-27

just graph these.

f) $y = -\frac{1}{2}x^2$



- 1) vertical stretch bafco $\frac{1}{2}$ and vertical reflection ($-\frac{y}{2}$)

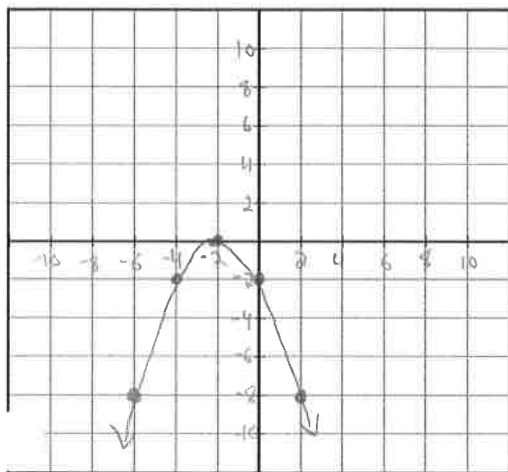
x	$-\frac{y}{2}$
-3	-4.5
-2	-2
-1	-0.5
0	0
1	-0.5
2	-2
3	-4.5

2) For each function $g(x)$:

$(-3, 9)$
 $(-2, 4)$
 $(-1, 1)$
 $(0, 0)$
 $(1, 1)$
 $(2, 4)$
 $(3, 9)$

- describe the transformations from the parent function $f(x) = x^2$
- create a table of values of image points for the transformed function
- graph the transformed function and write its equation

a) $g(x) = -2f\left[\frac{1}{2}(x+2)\right]$



- vertical stretch by 2; vertical reflection ($-2y$)
- horizontal stretch by 2 ($2x$)
- shift left 2 units ($x-2$)

$2x-2$	$-2y$
-8	-18
-6	-8
-4	-2
-2	0
0	-2
2	-8
4	-18

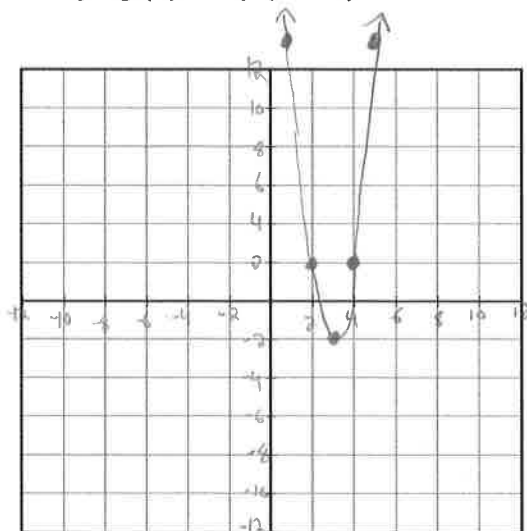
} graph these

$$g(x) = -2\left[\frac{1}{2}(x+2)\right]^2$$

$$g(x) = -2\left(\frac{1}{4}\right)(x+2)^2$$

$$g(x) = -\frac{1}{2}(x+2)^2$$

b) $g(x) = 4f(x-3) - 2$

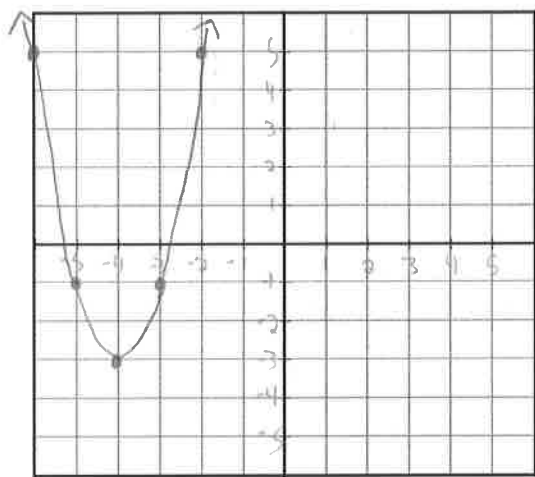


- vertical stretch by 4 ($4y$)
- shift right 3 units ($x+3$)
- shift down 2 units ($y-2$)

$x+3$	$4y-2$
0	34
1	14
2	2
3	-2
4	2
5	14
6	34

$$g(x) = 4(x-3)^2 - 2$$

c) $y = 2f(x+4) - 3$



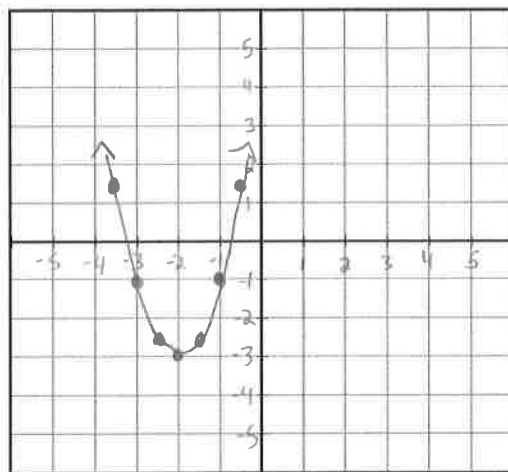
- 1) vertical stretch by 2 ($2y$)
- 2) shift left 4 units ($x-4$)
- 3) shift down 3 units ($y-3$)

$x-4$	$2y-3$
-7	15
-6	5
-5	-1
-4	-3
-3	-1
-2	5
-1	15

graph these

$$y = 2(x+4)^2 - 3$$

d) $y = \frac{1}{2}f[-2(x+2)] - 3$



- 1) vertical compression by $\frac{1}{2}$ ($\frac{y}{2}$)
- 2) horizontal compression by $\frac{1}{2}$; horizontal reflection ($\frac{x}{-2}$)
- 3) left 2 units ($x-2$)
- 4) down 3 units ($y-3$)

$\frac{x}{-2} - 2$	$\frac{y}{2} - 3$
-0.5	1.5
-1	-1
-1.5	-2.5
-2	-3
-2.5	-2.5
-3	-1
-3.5	1.5

$$g(x) = \frac{1}{2}[-2(x+2)]^2 - 3$$

$$g(x) = \frac{1}{2}(4)(x+2)^2 - 3$$

$$g(x) = 2(x+2)^2 - 3$$